

Section 1: Atmospheric Scrubbing Protocols

The habitat life support system relies on dual regenerative amine beds for carbon dioxide adsorption. Every 30 minutes, the desorbing bed is heated to 120 degrees Celsius under partial vacuum, releasing gaseous CO₂ into the Sabatier reactor feed line. Relative humidity is kept at 45 percent to avoid condensation in ventilation ducting.

Section 2: Emergency Depressurization Valves

In the event of rapid hull puncture, automatic isolation valves seal the affected module within 450 milliseconds. Crew members must deploy emergency mask quick-don units providing 100 percent oxygen at positive pressure. Airlock secondary bulkheads withstand differential pressure up to 120 kPa without structural deformation.

Section 3: Cryogenic Boil-Off Zero-Vent Management

Liquid methane and liquid oxygen tanks utilize 40 layers of multi-layer insulation and active pulse-tube cryocoolers operating at 20 Hz. Boil-off vapors are re-liquefied and returned to storage reservoirs, maintaining zero venting loss across the entire 500-sol expedition dwell period.